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COURSE: COMPUTER NETWORKS

LAB: 02

Task:01

Write down the advantages and disadvantages of RJ45 connectors.

Advantages:

- Standardized, reliable, and secure.
- Supports high speeds and low latency.
- Inexpensive and durable.

Disadvantages:

- Limits mobility (tethered by a cable).
- Creates physical clutter.
- Installation can be difficult; cables can be damaged.

Task:02

Briefly explain how the data is transmitted in wireless medium

1. Digital data is converted into a signal.
2. This signal is superimposed onto a radio wave (modulation).
3. The radio wave is transmitted through the air via an antenna.
4. A receiving antenna captures the wave.
5. The original data is extracted from the radio wave (demodulation).

Task:03

Briefly explain all type of network topologies.

- **Star:** All devices connect to a central hub (e.g., Wi-Fi router). Most common.
 - **Bus:** All devices connect to a single cable. Simple but outdated.
 - **Ring:** Devices are connected in a circle. Data travels in a loop.
 - **Mesh:** Devices have multiple connections for redundancy (e.g., the Internet).
 - **Tree:** A combination of star and bus topologies, like a hierarchy.
 - **Hybrid:** A mix of two or more different topologies.
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